

S'11: 7AN : CP407(1452)

SYSTEMS ANALYSIS AND DESIGN

Time : Three hours

Maximum Marks : 100

*Answer FIVE questions, taking ANY TWO from Group A,
ANY TWO from Group B and ALL from Group C.*

*All parts of a question (a,b,etc.) should
be answered at one place.*

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Group A

1. (a) Draw the schematic diagram of spiral model and explain the various phases involved in it . For which types of development projects is spiral model suitable ? 10
(b) Illustrate, with an example, the advantages of software prototyping. Why is it advisable to throw away the prototype in case of large scale software development ? 6 + 4
2. (a) Draw the context diagram and a set of data flow diagrams for developing a library management system with the following functionalities. List the assumptions made in your analysis. 12
(i) Inquiring the availability of a book by giving the ISBN or title or author's name

- (ii) Renting a book that is available
- (iii) Returning a book after use
- (iv) Reserving a book when it is not available
- (v) Collection of overdue charges
- (vi) Maintaining book details
- (vii) Maintaining supplier details who supply.
- (b) Briefly explain the Function Point Analysis method of software effort estimation. What is the advantage of using function point as compared to LOC (lines of code) for size estimation ? 8
- 3. (a) Show how the following relationships represented in E-R model with suitable examples : 12
 - (i) One-to-one
 - (ii) One-to many
 - (iii) Many-to-many
- (b) Summarize the advantages of using CASE tools in systems analysis and design. 8
- 4. (a) Assume that you are incharge of designing software to be installed in bank ATM. Which kind of architecture would you choose ? Give reasons for your selection and draw the architecture diagram of the system. 10
- (b) Draw flow-chart representation of branching and looping constructs. 5
- (c) What do you mean by a layered architecture ? What are the advantages of a layered architecture ? 5

Group B

- 5. (a) What are the advantages of modular design ? Bring out the role of coupling and cohesion in effective modular design. 12
- (b) What is document configuration ? Why is it needed ? 8
- 6. (a) What are the characteristics of a good test case ? 6
- (b) Write a program to identify the maximum value from a given set of integer values. Identify suitable test cases to test all possible paths of the program. 6
- (c) Distinguish between white-box and black-box testing methods. 8
- 7. (a) Develop a set of guidelines to conduct a program walkthrough. 8
- (b) Summarize the factors to be considered in forms design. 8
- (c) List the steps involved in dialogue design. 4
- 8. (a) Design a set of files for supporting a common payroll system. Comment on the data security issues of the file system you have designed. 6 + 4
- (b) Explain how management information systems help senior managers in decision making with an illustrative example. 10

Group C

- 9. Answer the following in brief : 10 × 2
 - (i) What are the limitations of the Waterfall model ?
 - (ii) Differentiate between technical feasibility and operational feasibility of a project.

- (iii) List *any two* common errors that are made while drawing data flow diagrams.
- (iv) What is a weak entity ? Give an example.
- (v) Why do we need to design software architecture ?
- (vi) Can a system be completely decoupled into independent modules ? Justify your answer.
- (vii) What is the purpose of system testing ?
- (viii) Who should be validating a system ? Why ?
- (ix) What are the major drawbacks of file system for storing large volumes of data pertaining to an application.
- (x) Give one example each for strategic, tactical and operational decisions.

W'12:7AN:CP407 (1452)**SYSTEMS ANALYSIS AND DESIGN***Time : Three hours**Maximum Marks : 100*

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Group A

1. (a) What is system ? Explain characteristics and types of system. 10
(b) Explain system development life cycle (SDLC) in detail.
How cost/benefit analysis plays an important role in SDLC ? 10
2. (a) What is the drawback of DFD ? Draw a DFD for the university admission system. 10
(b) Explain various types of feasibility studies that the analyst should consider. 10
3. (a) Write short notes on the following : 5 + 5
(i) System requirement specification
(ii) Data dictionaries.

- (b) Draw 0-level, 1-level and 2-level DFDs depicting various processes, data flow and data repositories for a retail readymade garment stores. Follow the appropriate conventions/symbols. 10

4. (a) What do you understand by system testing ? Explain different types of testing. 10
(b) Explain the cost-benefit analysis with an example. 10

Group B

5. (a) What are different methodologies of system design ? What do you understand by module coupling and cohesion ? 10
(b) What do you understand by quality assurance ? Explain goals of quality assurance. Discuss various methods of software testing. 10
6. (a) Explain the information support and nature of management for the following systems : 5 + 5
(i) Office Automation Systems
(ii) Transaction Processing Systems
(b) List and explain at least seven considerations for the input design. 10
7. (a) Explain the indexed-file organization method. List advantages and disadvantages in comparison with the sequential and random methods. 10
(b) Explain the following categories of project documentation : 5 + 5
(i) Operations documentation
(ii) User documentation.

8. (a) Explain various steps of the project review process. 10
(b) Discuss the design methodologies used in system design. 10

Group C

9. Answer the following in brief : 10 × 2
(i) Differentiate between static and dynamic system models.
(ii) What is meant by analyst/user interface ?
(iii) What is a decision tree ?
(iv) What is behavioural feasibility ?
(v) What is the significance of data dictionary ?
(vi) List characteristics of a good interface.
(vii) What is post-implementation ?
(viii) What is tailored software ?
(ix) Define module coupling.
(x) What is an activity network ?

W'13: 7AN:CP 407 (1452)**SYSTEMS ANALYSIS AND DESIGN**

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Group A

1. (a) Explain different types of CASE tools used in different phases of software life cycle. 7
- (b) Compare the relative advantages of a decision tree with a decision table for representing decision making. 6
- (c) Represent the different decision making that takes place when a user withdraws money from a bank ATM. 7
2. (a) Discuss the main activities carried out during feasibility analysis. 6
- (b) Explain the different ways in which an analyst gathers requirements from the customer. 6

- (c) Draw a labelled DFD of a Library Automation System. 8
3. (a) Explain the conceptual data modeling with entity relationship model. 6
- (b) Discuss the methods of designing interfaces and dialogues. 7
- (c) How will you design databases for an application? Explain with any one example. 7
4. (a) Draw ERD for a student information system for a college. Explain the concept of cardinality through it. 8
- (b) Explain the terms (i) Waterfall model, (ii) incremental model, and (iii) spiral process model. 3 × 4

Group B

5. (a) What is meant by cohesion and coupling in design solution? What is the relation between cohesion and coupling in modular design? Explain it with an example. 10
- (b) What is meant by system testing? Explain different types of system testing. 10
6. (a) What are the security issues in a computer system? How does an organization prevent its database from security concerns? Illustrate with an example. 10
- (b) Give steps in a form design. Explain at least five common GUI controls for inputs. 10
7. (a) What is MIS? What are different components of MIS? Explain different categories of information required to serve the needs of different levels of management in an organization. 10

- (b) What is DSS? Explain applications of DSS? What is the difference between MIS and DSS? 10
8. (a) Explain the following types of testing: (i) Alpha and Beta testing, (ii) recovery testing, (iii) security testing, (iv) stress testing, and (v) acceptance testing. 5 × 2
- (b) What are the important differences between code review and code walkthrough? Give examples of bugs identified by these two techniques. Briefly discuss how code review and code walkthrough are carried out. 10

Group C

9. Answer the following: 10 × 2
- (i) Why is code review considered a more efficient testing for bug reduction as compared to testing?
- (ii) Mention two main advantages of incremental process model.
- (iii) What is the main difference between analysis and design?
- (iv) What are basic elements of data modelling?
- (v) What is meant by the principle of "information hiding"? What is an advantage of using this principle?
- (vi) List the important activities in user interface design process.
- (vii) What is a program module? What is an advantage of a modular design?
- (viii) What are major drawbacks of file system?

- (ix) What is reverse engineering ? Why is it required ?
- (x) If a project is estimated to require 100 person-month of effort, does it mean that it can be completed by employing 100 software engineers for one month ? Explain briefly.

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Group A

1. (a) (i) Briefly describe the Waterfall model of software life-cycle with the help of a suitable schematic diagram. 6
- (ii) Discuss about its advantages and disadvantages. 4
- (iii) Give an example project for which Waterfall model is suited and an example project for which it is not suited. 2
- (b) What is prototyping ? Briefly explain the benefits of prototyping in software development. 8
2. (a) Distinguish between technical and economical feasibility studies. 6

- (b) Assume that you are given the following details of a small mail order catalogue system that allows people to shop from home :

When a customer receives the catalogue and wants to buy something, he can telephone, fax or email his order to the company. The company gets the order and sends the goods and an invoice. When the customer receives the goods with a delivery note, he sends payment and receives a receipt for payment. Draw the context diagram and data flow diagrams up to level 2 to analyze the requirements of this system. Also, give the data dictionary.

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3. (a) A university maintains data about the following entities with attributes :

(i) Course	Number, title, credits, syllabus and prerequisites
(ii) Course offered	Course number, year, semester, instructor, timings and classroom
(iii) Student	Student-id, name and degree program
(iv) Instructor	Identification number, name, department and title

The enrollment of students in courses and grades awarded to students in each course must be appropriately recorded. Construct an E-R diagram for the university using standard notation.

12

- (b) List and illustrate the primary uses and elements of a decision table.

8

4. (a) Illustrate the concepts of logical and physical design with a suitable example.

10

- (b) Develop architecture for a software system to be installed in the Automatic Teller Machine (ATM) of nationalized banks. Give justification for your selection. 10

Group B

5. (a) Explain different types of coupling and cohesion and discuss how these two factors influence functional independence. 12
- (b) Differentiate system simulation from system implementation with respect to objectives and methodology. 8
6. (a) How is a structured walkthrough conducted ? Provide a set of at least four guidelines for conducting such an event. 8
- (b) Illustrate the following testing strategies : 3 × 4
- (i) Unit testing
- (ii) Integration testing
- (iii) System testing
7. (a) What are the goals of input/output design ? 6
- (b) What are the characteristics of data entry ? Briefly explain how data entry requirements influence forms design. 8
- (c) List three heuristic rules for good dialogue design. 6
8. (a) Describe the sequential and indexed sequential organizations and comment on their applicability in file design. 10
- (b) Write short notes on the following :
- (i) Threats to system security
- (ii) Flow of data in Management Information Systems. 5 + 5

Group C

9. Answer the following in brief : 10 × 2
- (i) Under what circumstances can an analyst terminate a project ?
 - (ii) Suggest any two methods to overcome the difficulties in gathering accurate requirements from users.
 - (iii) What is a weak entity ? Give an example.
 - (iv) What is the difference between a flow chart and a data flow diagram ?
 - (v) Are there any limitations in structured program coding ? Give reasons for your answer.
 - (vi) Differentiate verification from validation.
 - (vii) What are the major advantages of database ?
 - (viii) Why do we need forms control ?
 - (ix) What are the major types of decisions made using a Management Information Systems ?
 - (x) List the limitations of password as a security mechanism.

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Group A

1. (a) Briefly explain how cost-benefit analysis is carried out during feasibility study. 8
(b) Briefly explain prototyping SDLC model with the help of a labelled process model. For which types of development projects is the prototyping model suitable. 12
2. (a) Suppose a tic-tac-toe game software has the following features : It first asks the user to enter his name and difficulty level for the game. The user and computer then play alternate moves until the game drawn or won by either. Based on the outcome, a congratulatory or consolation message for the user is displayed. Develop the level 0 and level 1 DFD for the tic-tac-toe game software. 12

- (b) Represent the decision taking involved consequent to each user entry of 'move' in the tic-tac-toe game in the form of a decision-table. 8
3. (a) What do you understand by the terms 'cohesion' and 'coupling' of a program module? How do these two concepts help arrive at good software design? 12
- (b) What do you understand by a layered design? What are the benefits of layering? 8
4. (a) Assume that using a grading software, a student registers for one or more courses in a semester and the corresponding teacher assigns a grade to the course at the end of the semester. Develop the ER diagram for this grading software. 12
- (b) Explain the difference between bottom-up and top-down software design. 8

Group B

5. (a) What do you understand by structured programming? What are its advantages? 10
- (b) What is the difference between code review and walkthrough? Explain your answer with respect to objectives and method for carrying out. 10
6. (a) Briefly explain how input-output design for a software can be carried out. 10
- (b) Identify the advantages and disadvantages of using file storage versus using DBMS. Give example situations where file storage may be preferable over using a DBMS. 10

7. (a) Differentiate between software verification and validation. 7
- (b) What are the differences between black-box and white box testings? 7
- (c) Design black-box test cases for a function named quad-solver (a, b, c) that takes three floating point numbers a, b and c representing a quadratic equation $ax^2 + bx + c = 0$ and displays its solution. 6
8. (a) Briefly explain the important components of an MIS. How are these components designed? 10
- (b) Differentiate among unit, integration and system testing. 10

Group C

9. Briefly answer the following: 10 × 2
- (i) What do you understand by phase containment of errors?
- (ii) What is the main disadvantage of classical life-cycle model?
- (iii) Give an example of a requirement problem that can be determined by requirement analysis.
- (iv) In which life-cycle phase is code review carried out?
- (v) Name one program verification technique and one validation testing.
- (vi) What do you mean by a program module?

- (vii) What problems may arise if a module has high coupling with other modules ?
- (viii) What problems may arise if a module has poor cohesion ?
- (ix) What is the objective of constructing a prototype while using a prototyping model ?
- (x) *Identify at least two aspects of software that are tested by performance testing.*

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1. (a) Describe the Waterfall model of software development. Discuss about the relative advantages and disadvantages of this model. Give two example projects for which this model is suitable. 10
- (b) Consider the cash flow of the project options shown in the table below : 10

Year	Project A, Rs. (in lakh)	Project B, Rs. (in lakh)
0	- 8000	- 8000
1	4000	1000
2	4000	2000
3	2000	4000
4	1000	3000
5	500	9000
6	500	- 6000

Calculate the net profit value for each of these projects with discount rate 8%. Which is the project more profitable ?

2. (a) Consider the development of an Employee Management System for an organization. The system is aimed at automating the complete employee details, including maintenance of employee attendance records, leave details, and printing monthly pay slips. This system is also expected to sum up each employee's work hours, time off, shift pays that help to generate the pay slip of employees. Draw the context diagram and data flow diagrams for developing this system. 10
- (b) Illustrate the usage of decision tables in describing a process with an example. 10
3. (a) A collection of information is to be recorded in a company's personal database : 12
 - (i) The company has a set of departments.
 - (ii) Each department has a set of employees, a set of projects and set of offices.
 - (iii) Each employee has a job history (set of jobs the employee has held). For each such job, the employee also has a salary history (set of salaries received while employed in that job)
 - (iv) Each office has a set of phones.

The database is to contain the following information :

 - **For each department** : Department number (unique), budget and the department manager's employee number (unique).
 - **For each employee** : Employee number (unique), current project number, office number and phone

number. Also, title of each job the employee has held and the date and each distinct salary received in that job.

- **For each project** : Project number (unique) and budget.
- **For each office** : Office number (unique), area and phone numbers.

Develop an E-R diagram for this database.

- (b) Summarize the usage of CASE tools in structured systems analysis. 8
4. (a) Assume that you are incharge of designing software to be installed in bank ATM. Which kind of architecture would you choose ? Give reasons for your selection and draw the architecture diagram of the system. 10
- (b) Represent with flow-charts how *branching* and *looping* constructs are accommodated in a structured program. 10

Group B

5. (a) What are the advantages of modular design ? Bring out the role of coupling and cohesion in effective modular design. 12
- (b) What is document configuration ? Why is it needed ? 8
6. (a) Distinguish between white-box and black-box testing methods with an example. 8
- (b) Briefly explain unit testing, integration testing and system testing strategies. Also, list the usefulness and limitations of these strategies. 12
7. (a) Develop a set of guidelines to conduct a program walkthrough. 8

- (b) Summarize the factors to be considered in forms design. 8
- (c) List the steps involved in dialogue design. 4
8. (a) Design a set of files for supporting a common payroll system. Comment on the data security issues of the file system you have designed. 6 + 4
- (b) With an illustrative example, explain how management information systems help senior managers in decision-making. 10

Group C

9. Give brief answer to the following : 10 × 2
- (i) What are the advantages of software prototyping ?
- (ii) With a simple example, illustrate how the Return on Investment is computed for a project.
- (iii) How are flow-charts different from data flow diagrams ?
- (iv) Give an example for many-to-many relation. How is this resolved in Entity-Relationship diagram ?
- (v) List the advantages of layered software architecture.
- (vi) Define *temporal cohesion*. Give an example situation where this concept will be useful during modular programming.
- (vii) Why do we need regression testing ?
- (viii) Distinguish between verification and validation.
- (ix) What is the purpose of overflow area in indexed sequential access method ?
- (x) Suggest any *two* methods to enhance access security provided by passwords.

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Group A

1. (a) Draw 0-level, 1-level and 2-level DFDs depicting various processes, data flow and data repositories with an appropriate example. Also, follow the appropriate conventions/symbols. 10
- (b) What do you understand by cost estimation? Briefly explain how cost estimation for a project can be achieved. Schematically draw the classical life cycle model. 10
2. (a) Identify various outputs/products/reports associated with each and every phase of SDLC. 5
- (b) Explain what is modularity. How can modularity of a program be determined? 7

- (c) What do you mean by requirements analysis ? What is its objective ? How is requirements analysis performed ? 8
3. (a) Write short notes on the following : 3 + 3 + 4
- (i) Decision Support System
 - (ii) CASE tools
 - (iii) Software testing
- (b) What is an entity-relationship (ER) diagram ? How is it useful in program development ? Give the ER diagram of a simple library information system. 10
4. (a) What do you understand by feasibility study ? Explain how feasibility study is carried out with an example. 8
- (b) Write short notes on the following : 4 × 3
- (i) DFD
 - (ii) Software architecture
 - (iii) Decision table
 - (iv) Data dictionary
- Group B**
5. (a) What is MIS ? Explain the role of MIS in a business organisation. Also, explain how MIS can be used for management control and strategic planning of an organisation. 12
- (b) What is system design ? Explain top-down design with an example. 8
6. (a) Briefly explain what you understand by cohesion and coupling. How can cohesion and coupling of a given design be determined ? 10
- (b) Explain the differences between review and walk-through. Briefly explain how these are carried out and the types of bugs detected by these two techniques. 10
7. (a) Differentiate between verification and validation. Explain how verification and validation are carried out. 10
- (b) What are the different documents that are produced during software development ? Distinguish between internal and external documentation. Discuss the ways internal documentation is carried out. 10
8. (a) What is prototype of system ? What are its advantages and what are the steps required to build a prototype ? 8
- (b) What do you mean by cost and benefit analysis ? Illustrate cost-benefit analysis using an example. 6
- (c) Briefly discuss how dialog design can be carried out. 6
- Group C**
9. Answer the following in brief : 10 × 2
- (i) Write the role of a system analyst.
 - (ii) Briefly explain how gathering of requirements can be carried out.
 - (iii) What is structured analysis ? Name a tool to carry out structured analysis.
 - (iv) What are different types of system testing ? Write who carries out each.
 - (v) What is a layered design ? What are its advantages ?

- (vi) Differentiate between black box testing and white box testing. Can one be used in place of another.
- (vii) What is system simulation ? How does system simulation help in program development ?
- (viii) What do you mean by system maintenance ? Explain with the help of a suitable example.
- (ix) What is a form ? Mention various types of forms.
- (x) What are the different security threats to an MIS application ? Briefly explain.

S'16: 7 AN : CP 407 (1452)

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Group A

1. (a) Briefly explain different stages in System Development Life Cycle. How are these stages included in spiral model of software development ? 10
(b) Describe various methods used in fact finding. 10
2. (a) Illustrate the evaluation methods that are available to evaluate an identified financial data. 10
(b) How can requirements gathered be validated using prototyping ? Discuss about the limitations of this methodology. 10
3. (a) Distinguish between flow-chart and data flow diagrams. 8

(Turn Over)

- (b) Draw context diagram, level-1, level-2 DFDs for a standard Hospital Management System. 12

4. (a) What is the network architecture used in E-commerce? Explain briefly how this architecture works. 10
- (b) Write short notes on the following : 10
- (i) Parameter passing
- (ii) Execution sequence

Group B

5. (a) Elaborate on any two CASE tools used in designing a software. 10
- (b) Briefly explain the role of coupling and cohesion in effective modular design. 10
6. (a) Compare different types of online user dialogs for transaction processing and discuss about their advantages. 8
- (b) Distinguish between verification and validation with a suitable example. Can one be used in place of the other? Explain. 12
7. (a) What are the major objectives of a walkthrough? List the guidelines for conducting a walkthrough. 8
- (b) Briefly explain *three* major file organizations highlighting their advantages and applicability. 12
8. (a) Explain in detail about the input design process. 10
- (b) Differentiate between corrective, perfective and adaptive maintenance of software. Discuss the process models for carrying out software maintenance. 10

S'16:7AN:CP407 (1452) (2)

(Continued)

Group C

9. Briefly answer the following : 10 × 2
- (i) State how Alpha testing is different from acceptance testing.
- (ii) Identify two problems that may arise, if the SRS document is written as a narrative essay.
- (iii) How do tangible costs differ from direct costs?
- (iv) How decision table overcomes the drawbacks of a decision tree?
- (v) Differentiate between error tolerance and error avoidance.
- (vi) List the limitations of unit testing.
- (vii) Identify two differences between white box and black box testing of a software.
- (viii) Why are user interface tools built up in layers?
- (ix) Differentiate between horizontal integration and vertical integration of CASE tools.
- (x) Differentiate between iconic, command-driven and menu-driven user interfaces.

S'16:7AN:CP407 (1452) (3)

AG-800

W'16: 7 AN : CP 407 (1452)

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Group A

1. (a) What are the different phases of systems Development Life Cycle ? Explain each phase briefly. 10
(b) What are the system requirements and how to validate the system requirement ? 10
2. (a) Explain the different techniques of information gathering. 10
(b) Explain the use of sequence diagram with a suitable example. 10
3. (a) What is Entity-relationship diagram ? Draw ER-Diagram for the following scenario : 10

(Turn Over)

A company has several departments. Each department has a supervisor and at least one employee. Employees must be assigned to at least one, but possibly more departments. At least one employee is assigned to project, but an employee may be on vacation and not assigned to any projects. (Assume suitable entity types and attributes to form ER-Diagram)

(b) Name and describe the relationship in a Use Case diagram ? 10

4. (a) What are the common factors for feasibility study ? Explain them in detail. 10
- (b) Check the following PERT Network diagram for a project. The activities time line costs are such that the crash cost of node 3 is 4 and crash cost of node 6 is 3. Complete the PERT sequence with their prediction cost. 10

Group B

5. (a) What is the need of modularizing the system and explain the following cohesion techniques with example : 12
- (i) Logical Cohesion
- (ii) Temporal Cohesion
- (iii) Procedural Cohesion.
- (b) What is the need to write a design document and how to write a good design document ? 8
6. (a) How we differentiate between verification and validation ? Write five difference. 10
- (b) Consider the following code. Draw the data flow graph and find the cyclomatic complexity : 10

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(Continued)

```
i = 0;
n = 4; //N-Number of nodes present in the graph
while (i < n - 1) do
    j = i + 1;
    while (j < n) do
        if A[i] < A[j] then
            swap (A [i], A[j]);
        end do;
        i = i + 1;
    end do;
```

7. (a) What is user interface ? How we design a user interface, explain in detail. 10
- (b) What guidelines should be analysts follow in the design of input and output ? 10
8. (a) How we implement security and control in the design of an information system ? 10
- (b) How MIS is useful now a days and what are the different types of MIS ? 10

Group C

9. Briefly answer the following questions : 10 × 2
- (i) Define the contest coupling.
- (ii) Differentiate between Quality assurance and Quality Control.
- (iii) Explain the terms Refinement with the help of example.
- (iv) Explain the terms Abstraction with the help of example.
- (v) Justify the statement that system development is a process of progressive change.

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(Turn Over)

- (vi) What are the reasons for changes in requirements ?
- (vii) What do you mean by crashing ?
- (viii) What are functional requirements and non-functional requirements ?
- (ix) What is waterfall model its advantages and disadvantages ?
- (x) What is the difference between function oriented and object oriented design ?

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